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Title: Management of Congenital Cytomegalovirus (CMV)

Management of Congenital Cytomegalovirus (CMV)Infection guideline

Introduction:

Cytomegalovirus (CMV) is the most common viral cause of congenital infection, affecting 0.2–2.2% of all livebirths. Around 10–15% of neonates with congenital CMV will be symptomatic at birth. ^[1]. It is the leading non-genetic cause of sensorineural hearing loss (SNHL) and a major cause of neurological disability.

Death attributable to congenital CMV is estimated to occur in 3% - 10% of infants with symptomatic disease, or 0.3% to 1% of all infants with congenital CMV infection. Among asymptomatic neonates, 5 -15%eventually develop neurologic sequelae; hearing loss is the most common. Early treatment is critical to prevent or reduce progression of SNHL in a child with congenital CMV infection. Approximately 20% of all hearing loss at birth and25% of all hearing loss at 4 years of age is attributable to congenital CMV infection.

Objectives:

-To provide a clear guideline for management and follow up of congenital CMV.

Symptoms and signs: In less than 10% of congenital CMV

- Rash
- Jaundice
- Microcephaly
- Low birth weight
- Intrauterine growth restriction
- Hepatosplenomegaly
- Seizures
- Retinitis

Clinically Detectable Symptoms/Signs

Physical examination

Small for gestational age (birth weight ≤ 2 standard deviations for gestational age)
 Microcephaly (head circumference ≤ 2 standard deviations for gestational age)
 Petechiae or purpura (usually found within hours of birth and persist for several weeks)
 Blueberry muffin rash (intradermal hematopoiesis)
 Jaundice *
 Hepatomegaly
 Splenomegaly

Neurologic physical examination

Microcephaly (head circumference ≤ 2 standard deviations for gestational age)
 Neurologic signs (lethargy, hypotonia, seizures, poor sucking reflex)

Abnormalities Detected Incidentally or Through Subsequent Investigation/Specialist Examination

Laboratory results

Anemia
 Thrombocytopenia (occurs in the first week, but platelets often increase spontaneously after the second week)
 Leukopenia, isolated neutropenia
 Elevated liver enzymes (alanine aminotransferase/aspartate aminotransferase)
 Conjugated hyperbilirubinemia

Cerebrospinal fluid

Abnormal cerebral fluid indices, positive CMV DNA

Neuroimaging

Calcifications, periventricular cysts, ventricular dilatation, subependymal, pseudocysts, germinolytic cysts, white matter abnormalities, cortical atrophy, migration disorders, cerebellar hypoplasia, lenticulostriatal vasculopathy

Hearing test

Sensorineural hearing loss uni- or bilaterally

Visual examination

Chorioretinitis, retinal hemorrhage, optic atrophy, strabismus, cataracts

* CMV-associated jaundice can be present on the first day after birth and usually persists longer than physiologic jaundice.

-Diagnosis:

1-Blood CMV PCR.

2-Urine CMV PCR in the 1st3 weeks of age.

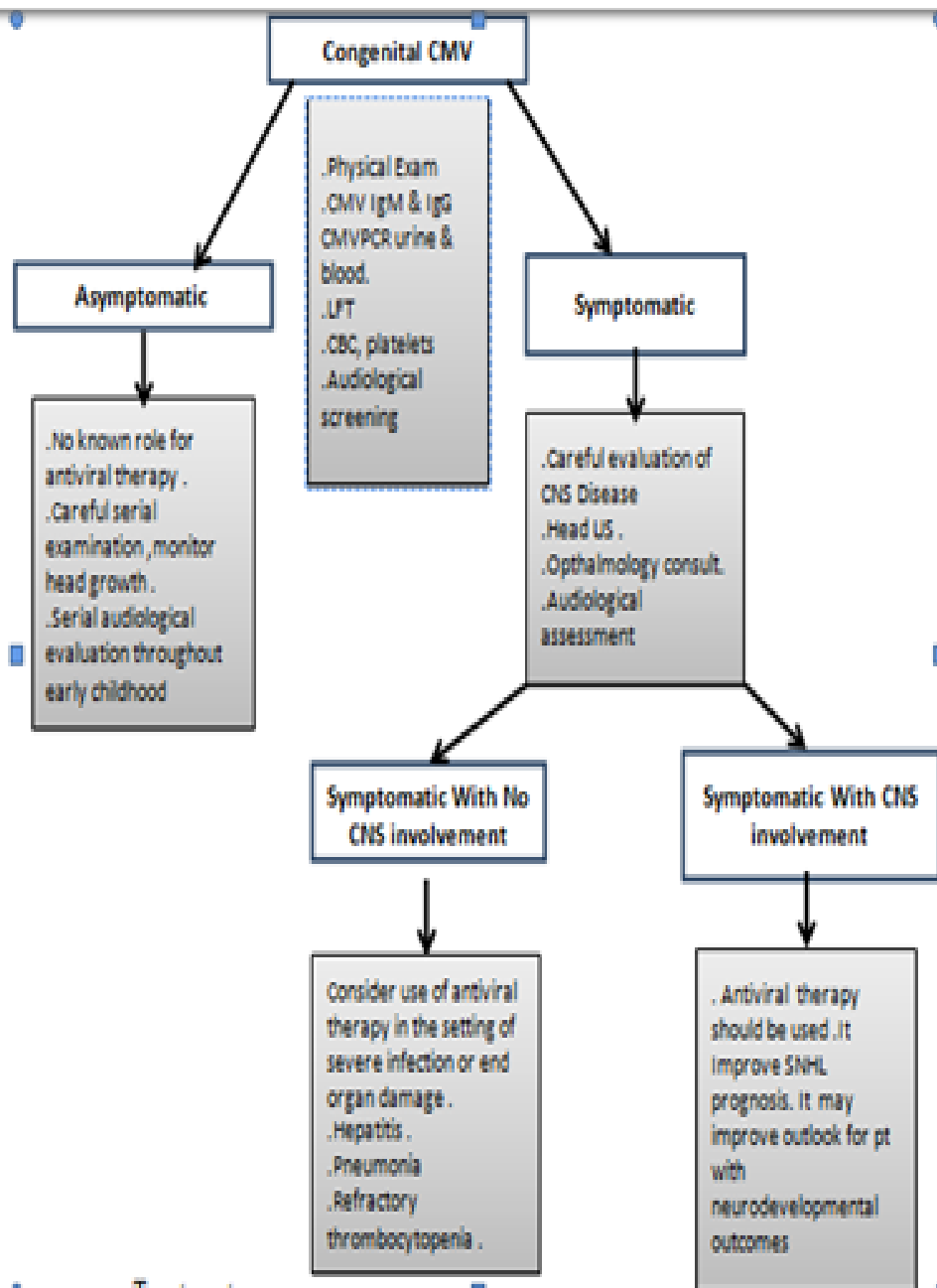
3-Saliva PCR can be used

3- Serology: (if the urine CMV sent after 21days of life)

- IgG avidity assays distinguish primary from non-primary CMV infection

- Avidity increases over time reflecting maturation of the immune response





Treatment:

Treatment:

- Any infants with evidence of CNS involvement
- Acutely unwell infants with severe focal or multisystem disease
- Treatment help prevent hearing loss & developmental delay in infants with symptomatic infection at 12 and 24 months of age.

Doses and Duration:

Oral valganciclovir is first line treatment and IV ganciclovir should only be used if oral medication is not tolerated or if there is severe disease and absorption is uncertain

Valganciclovir oral 16mg/kg/dose twice daily for 6 months.

OR

OR

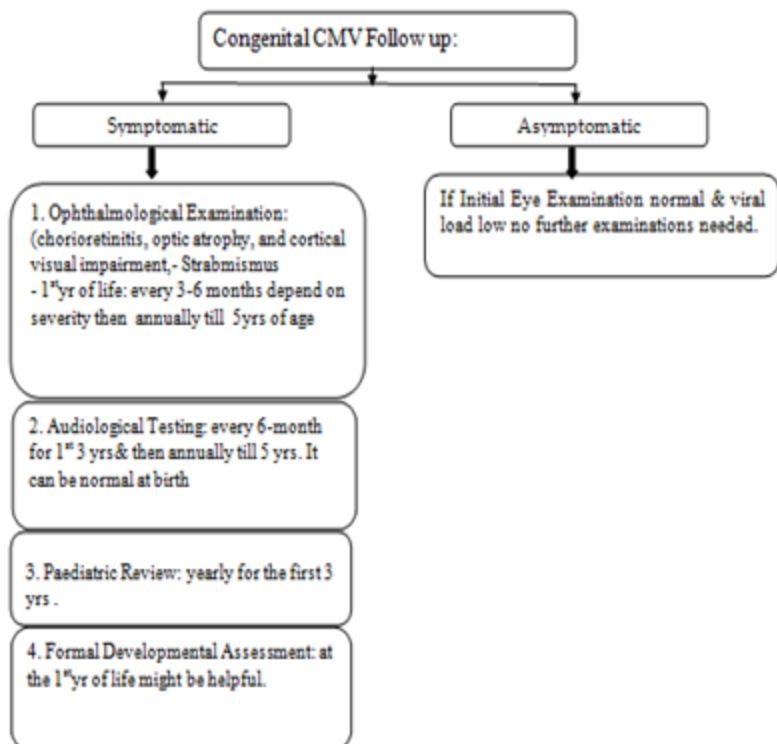
Ganciclovir at initiation followed by oral Valganciclovir (on advice of pediatric Infectious Diseases)

- Firstline treatment is now preferentially oral valganciclovir unless severe symptomatic CNS disease or the baby is unable to tolerate oral medication.

Prevention of CMV acquisition:

Handwashing after contact with urine or saliva, and avoiding sharing utensils, drinks or food with young children.

Congenital CMV Follow up:



FBC, LFTs Weekly for 6 wks, then at 8 weeks, then monthly till end of therapy testing:

Neutrophil Count (N):
consider GSCF for
neutropenia ($<0.5 \times 10^9/L$)
- If $N=0.2 \times 10^9/L$; stop
until $>0.5 \times 10^9/L$
- If $N=0.2-0.5 \times 10^9/L$ then
recheck weekly &
consider stopping if
persistent

Platelet: If $plts < 20 \times 10^9/L$
stop until $> 50 \times 10^9/L$ then
restart.

Renal Function Test: If
creatinine increase adjust
to renal dose

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
1-Congenital Cytomegalovirus Infection: Update on Treatment. Royal College of Obstetricians & Gynecologists C.	Khalil A, Heat P, Jones C	Nov 2017	1-3
2-Congenital Cytomegalovirus Infection: New Prospects for Prevention and Therapy <i>for</i> Pediatric Clinics of North America: Advances in Evaluation, Diagnosis and Treatment of Pediatric Infectious Disease	Elizabeth C.Swanson, DO and Marke R.Schleiss	Apr 2013	1-2
3-Congenital and Perinatal Cytomegalovirus Infection (CMV),. Treatment Guideline for Infants with Congenital CMV Disease	By Mary T. Caserta	October 2015	1-6
4- Chiopris G, Veronese P, Cusenza F, Procaccianti M, Perrone S, Daccò V, Colombo C, Esposito S. Congenital Cytomegalovirus Infection: Update on Diagnosis and Treatment. <i>Microorganisms</i> . 2020 Oct 1;8(10):1516. doi: 10.3390/microorganisms8101516. PMID: 33019752; PMCID: PMC7599523.	Chiopris G, Veronese P, Cusenza F, Procaccianti M, Perrone S, Daccò V, Colombo C, Esposito S.	2020	919-935.
5. Cytomegalovirus infection,	Report of the committee on infectious diseases	2018	1-3

guidelines/nhsggc-guidelines/neonatology/cytomegalovirus-cmv-congenital-infection/

NHSGGC Paediatrics for
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Review Details

	Review BY	Review Date
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